BIOL 3712: Molecular Biology

Not an official Northeastern University syllabus. This course is invented for test volume and is not in the institution's catalog.

Credits	5 semester credits	**Term**	Fall 2024
Contact hours	75 hours (lecture + lab)	**Instructor**	Synthetic Instructor 11
Prerequisites	One introductory course in the discipline	**Office hours**	(not applicable - test file)

Course description

Molecular Biology. Topics include laboratory techniques, transcription, PCR, recombinant DNA and genomics; later weeks address gene regulation, translation and DNA replication.

Learning outcomes

By the end of the course, students will be able to:
1. Explain the core ideas of laboratory techniques.
2. Apply transcription to solve standard problems.
3. Analyze problems involving PCR.
4. Use recombinant DNA in practical exercises.
5. Evaluate approaches to genomics.
6. Communicate findings about gene regulation.

Weekly schedule

Week	Topic	Activity / assessment
1	Laboratory techniques	Problem set
2	Laboratory techniques (continued)	Lab or workshop
3	Transcription	Quiz
4	PCR	Case study
5	PCR (continued)	Problem set
6	Recombinant DNA	Reading and discussion
7	Review and midterm	Midterm exam
8	Genomics	Lab or workshop
9	Genomics (continued)	Quiz
10	Gene regulation	Case study
11	Translation	Problem set
12	Translation (continued)	Reading and discussion
13	DNA replication	Problem set
14	Review and synthesis	Final exam / project

Assessment

Component	Weight
Problem sets / assignments	25%
Labs	15%
Midterm exam	25%
Final exam or project	30%
Participation	5%

Materials and policies

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Required materials: instructor-provided open course notes and readings posted on the course site. Grading scale: A 93-100, A- 90-92, B+ 87-89, B 83-86, B- 80-82, C+ 77-79, C 73-76, C- 70-72, D 60-69, F below 60. Policies: see the institution's own catalog; none are reproduced here.

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